

RANGA SAI PRASANTH MANDA

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SUMMARY

Software Development Engineer proficient in building large-scale systems and experienced with leading projects that enhance performance and customer experience. Previously worked on microservices scaling, deployment automation, and UI optimization, integrating machine learning models for business solutions. Key accomplishments include designing event-driven services handling over 100k daily events and improving backend performance for 50,000+ users. Aspires to apply expertise in software engineering to build advanced solutions for large scale systems, delivering enhanced capabilities to customers.

EDUCATION

University at Buffalo - SUNY

December 2024

Master's, Computer Science

GPA: 4.0

- Coursework : Computer Security, Operating Systems, Data Intensive Computing, Data Models & Query Languages, Algorithm Design & Analysis, Machine Learning, Deep Learning, Computer Vision & Image processing.

National Institute of Technology (NIT) Calicut

June 2020

Bachelor's, Electronics and Communication Engineering

GPA: 3.7

- Coursework : Data Structures & Algorithms, Computer Architecture, Computer Networks, Machine learning, Artificial Intelligence.

PROFESSIONAL EXPERIENCE

University at Buffalo, The State University of New York

Buffalo, NY, USA

Graduate Teaching Assistant

August 2024 – Dec 2024

- Worked as a Teaching Assistant for **Data Models and Query Languages** course, guiding students on **Relational Databases**.
- Mentored** a group of 20 students on their course project and conducting office hours to help students clear their doubts.

RadicalAI

New York, NY, USA

Software Engineer intern

May 2024 - August 2024

- Refined the UI of Kai, a teaching assistant chatbot powered by **OpenAI's GPT-4.5**, using **React**, **JavaScript**, **Figma** and **Material-UI**.
- Enhanced backend performance with **Node.js** and **Google Firebase**, leading to improved user experience for 2,000 users.

Deutsche Bank

India

Software Development Engineer II | Associate | Risk Finance Treasure Division

August 2022 - August 2023

- Designed **Scalable, event-driven** Microservices for ETA(Estimated Time of Arrival) Tool leveraging **Python** and **Flask**, handling **data processing** and **state management** of 100k+ events daily for a Profit and Loss (PnL) Reporting Platform.
- Implemented **RESTful APIs** for seamless integration between microservices, Database, Splunk and frontend applications, ensuring scalability, security, and ease of use.
- Crafted a **regression** model to predict ETA of **compute intensive** PnL calculations, saving 2+ hrs of L2 support daily.
- Wrote a package for **JWT** based **authentication** (Client Credentials and Authorization code flow) using **OAuth 2.0**
- Set up a **Jenkins CI/CD** pipeline to build, deploy and manage **Python container** images on **OpenShift** leveraging **Docker**, **Helm** and **Kubernetes**, reducing deployment time by 50% while enhancing reliability and automating scaling.
- Effectively managed all phases of the **software development life cycle**, including coding standards, code reviews, source control management, build processes, testing, and operations.
- Adhered to **Agile** methodologies and participated in **Scrum** ceremonies to ensure efficient project delivery.

Deutsche Bank

India

Software Engineer | Senior Technology Analyst | Private Bank Wealth Management Division

July 2020 - July 2022

- Built and optimized **microservices** for financial reporting platforms, improving performance and scalability.
- Revamped a **Java Spring Boot Microservice** and a **React** dashboard to run PnL Dataflows, lowering data processing time.
- Engineered a data pipeline to transfer event logs from **Splunk** to **PostgreSQL**, improving data accessibility.
- Deployed Open-Source **PostgreSQL** high throughput database fully on-prem using **docker** and **Openshift** Cloud and automated partitions for 10TB of data, ensuring high availability, fault tolerance, scalability and enhanced data reliability.
- Developed an "Automated login trend reporting" tool with **Python** and **pandas**, decreasing manual reporting effort by 60%.
- Developed **project business cases**, analyzed 30+ **JIRA stories** and collaborated with 5+ cross-functional teams, delivered 10+ features through effective **stakeholder engagement** and requirement documentation.

PROJECTS & OUTSIDE EXPERIENCE

AI-Powered Chat Application with LLaMA 2 Using Ollama and Spring Boot

August 2024 - Present

- Developed a local AI-powered chat application using Meta's LLaMA 2 model with Spring Boot and Java, enabling real-time user interactions and customizable prompt settings.
- Integrated Ollama to run and optimize LLaMA 2 models locally, ensuring low-latency responses without cloud dependency.
- Deployed chat application logs and configurations to Amazon S3, enhancing data accessibility and durability.
- Key Skills: Java, Spring AI, Spring Boot, Ollama, Meta's LLaMA 2, RESTful APIs, AWS S3, Docker, Maven.

MemDBLite: In-Memory Key-Value Database in C++

January 2024 - May 2024

- Implemented a in-memory key-value database with approximate LRU eviction policy using the Clock algorithm, ensuring optimal memory usage while maintaining high performance for the database
- [Link to project](#)

Healthcare Database Management System

January 2024 - May 2024

- Configured a PostgreSQL healthcare database with solid design principles for data management of 50,000 users.
- Designed an engaging web application using Python, Flask and HTML, enabling user interaction.

Pintos Operating System (C,OS,Git,Docker,Advanced System-Level Programming)

January 2024 - May 2024

- Implemented advanced threading, synchronization, priority donation, MLFQS, safe memory access, system calls for Pintos operating system using C, reducing context switch latency by 20%.

CDN Caching System

January 2024 - May 2024

- Developed a high-performance distributed CDN caching system using Go, gRPC, and Redis, incorporating data partitioning, replication, and eviction policies to efficiently cache frequently accessed data across multiple nodes.

Advanced Text Processor application (Java , Python,Linux , Distributed systems)

August 2023 - December 2023

- Built a text analytics solution using Java, Hadoop MapReduce and PySpark, leveraging HDFS for distributed storage, robust batch processing, and PySpark for fast, in-memory processing, enhancing speed and accuracy

Bank Term-Deposit Predictor Application

August 2023 - December 2023

- Developed a predictive ML model and a web application utilizing ML, Python, Flask, HTML, CSS for term deposit prediction and trend analysis of a Portuguese Bank campaign, achieving a 15% increase in prediction accuracy
- [Link to project](#)

Jasper-smart-robot

July 2019 - November 2019

- Developed a robot that recognizes faces based on haarcascade algorithm and interacts with people using google speech to text and text to speech engines. Deployed it on edge device Raspberry Pi integrated with a camera module.
- Key Skills: OpenCV, Python, Speech Recognition, Image Processing, Object detection,Face Recognition,Robotics.
- [Link to project](#)

SKILLS

Programming Languages: Python, Java,C++,C, Javascript,JS,Typescript, Go,Bash,HTML/CSS,PHP,SQL.

Frameworks & Libraries: Spring Boot, React.js, Flask,Django, FastAPI, Node.js, Express.js, Redux.js, MERN.

Web Development: HTML, CSS,PHP, Webpack, REST APIs, web services , JSON, frontend ,backend,Webkit.

Databases: PostgreSQL, MySQL, Oracle, NoSQL,MongoDB, Cassandra, Redis,DynamoDB, Redshift.

Cloud : AWS,GCP,RDS, S3, EC2, Lambda, CloudFormation, IAM, VPC,Redis, DynamoDB, CloudWatch, Route 53.

DevOps: Docker, Kubernetes, OpenShift, Jenkins, Maven, Gradle, Helm , CI/CD.

Tools & Platforms: Git, GitHub, Bitbucket, JIRA, Postman, Grafana, Prometheus, Confluence.

Networking Protocols: HTTP, HTTPS, TCP/IP, UDP, FTP, SMTP, DNS, DHCP, SNMP.

Software Development: Microservices, large-scale systems, system design, unit test, MVC, TDD, Agile, kanban, SDLC

Data Engineering & Distributed Systems: Hadoop, Apache Kafka, Splunk, Cassandra

Data Analysis & Machine Learning: TensorFlow, Keras, PyTorch ,NumPy, Pandas, Apache Spark, NLP

Other Skills: Debugging, Performance optimization, IAM, Customer Service, Quality Assurance, Test Engineering, unit tests, Project management, troubleshooting , communication skills , test driven development,BDD,

CERTIFICATIONS

[AWS Solutions Architect - Associate](#) [Full Stack Web Development Internshala](#) [Associate Cloud Engineer GCP.](#)

[Ongoing: IBM Full Stack Software Developer] [C++] [Java] [Python] [JavaScript] [SQL] [Data Visualization with PowerBI]